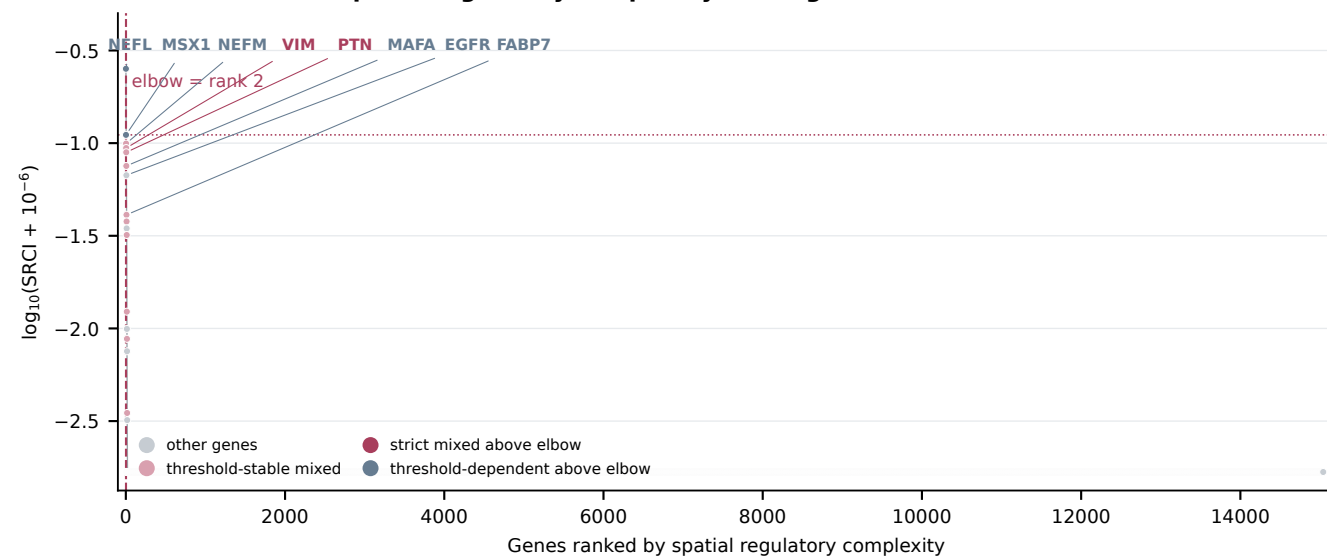
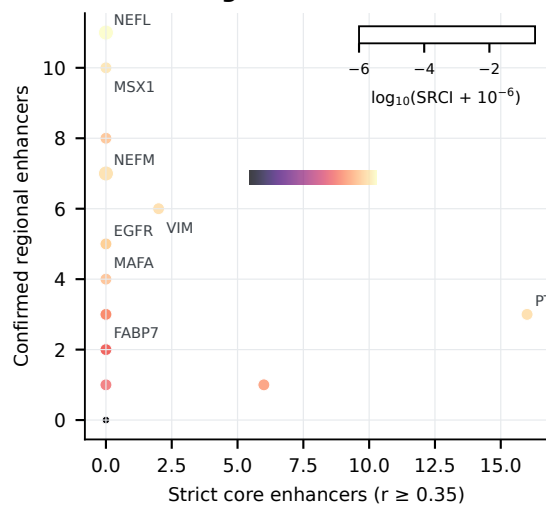


GW12 spatial regulatory complexity landscape

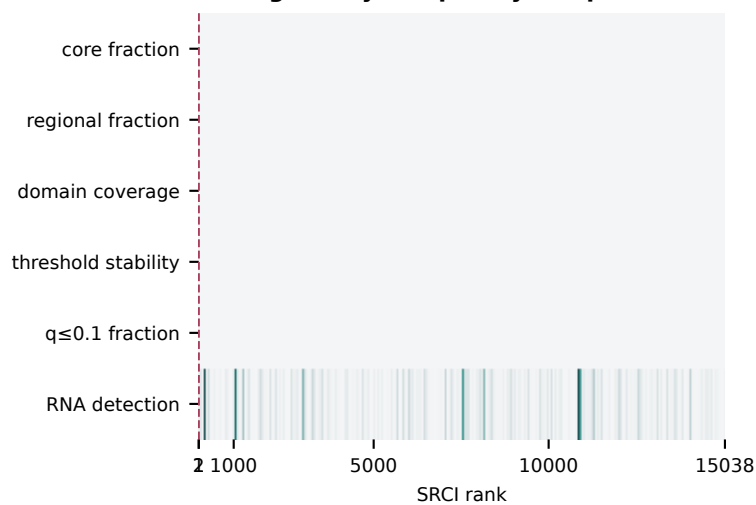
a Genome-wide spatial regulatory complexity ranking



b Core-regional architecture



c Regulatory-complexity composition



SRCI: mean mixed-regulation score across core thresholds 0.20/0.25/0.30/0.35. All potential non-zero candidates were evaluated using 99 two-dimensional spatial shifts; regional enhancers require local $r \geq 0.20$, evidence ≥ 0.005 and within-gene BH $q \leq 0.10$.